Cost-Aware Multi-Model Inference Router (CAMIR)

Project Proposal

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Abstract

Routing and cascading send easy prompts to a small language model and hard ones to a large model. CAMIR builds and evaluates such a router on a single Apple M4 laptop, with local small and medium tiers and a hosted large tier. The project makes four contributions: (1) a measured account of the *cost of escalation itself*—the cost, delay and energy wasted when a small-model attempt is rejected—and policies that reduce it; (2) a new family of escalation *confidence signals*, including signals read from a mixture-of-experts router; (3) *calibrated routing with guarantees*, a risk-controlled threshold that bounds the quality lost relative to always using the large model, for policies that skip or abort the small attempt; and (4) *per-step routing in agentic workflows*, where switching tiers mid-trajectory carries its own cost. For each contribution we state what prior work already establishes and what remains open.

1 Introduction

Large hosted models are accurate but costly and slow; small local models are cheap and fast but fail more often. Pre-generation routers [2, 4] and confidence-gated cascades [1, 5, 6, 8] try to capture the savings, and cascade routing [3] combines the two. Vendors now ship routers as products [21, 22, 23, 24]. The project runs on a MacBook M4 with a pay-per-token API for the large tier: no HPC or cloud credits are needed.

2 Problem Statement

Given a stream of prompts (and, in the agentic setting, multi-step tasks) and a pool of tiers (small, medium, large), choose for each prompt or step which tier answers, so that quality stays within ϵ of always using the large tier while cost, latency (mean and tail) and energy are minimized, and so that the quality statement is backed by a statistical guarantee rather than a tuned threshold. Published evaluations mostly report token cost or the share of calls sent to the large model [1, 2, 7]; surveys note the lack of shared metrics, pools and cost bases [17] and of non-financial costs [18].

3 Contributions and Novelty

Honest positioning. Several of these topics are already active. Bouchard [7] characterizes threshold cascades decision-theoretically and finds that a pre-generation router often beats the

best cascade, attributed mainly to avoiding the cheap model's cost. Conformal methods for routing and cascading already exist [9, 10, 11]. Budget-aware per-step routing for agents exists [13] (and [14], which we have not yet read). We therefore state, for each contribution, a narrower gap that we have not found covered in the sources we read; our literature search is not exhaustive and each claim is to be re-checked with the advisor before submission.

3.1 C1: The cost of escalation itself

Established. A cascade pays for every model it queries [1, 10], and the cheap model's cost explains much of the router's advantage in token cost [7].

Our contribution. With c_S, c_L the tier costs, g the gate cost, e the escalation rate and $\rho = (c_S + g)/c_L$, a cascade saves

$$\text{Savings} = (1 - e)\Delta - eW,$$
$$\Delta = \mathbb{E}[c_L - c_S - g \mid \text{acc}], \quad W = \mathbb{E}[c_S + g \mid \text{esc}],$$

breaking even at $e^* = \Delta/(\Delta + W)$ ($= 1 - \rho$ for homogeneous costs). Per prompt, attempt the small model first only if the predicted acceptance probability $\hat{a}(x) > \rho(x)$. For latency, escalated requests pay $t_S + t_g + t_L$, so under idealized assumptions a cascade's p95 beats always-large only if $e < 5\%$. These relations are elementary; the contribution is (i) measuring the tax in dollars, seconds (mean and tail) and joules on real hardware with a local-plus-hosted pool, and (ii) two policies built from it: **P3**, a skip-rule route-then-cascade, and **P4**, an early-abort cascade that abandons the small attempt when the running mean log-probability falls below a threshold after n_0 tokens.

3.2 C2: A new confidence signal

Established. Token log-probability, top-1/top-2 margin (with isotonic calibration [8]), sampled-answer agreement [19, 6], self-verification [5], and probes on hidden states or prompt embeddings are used to decide escalation.

Gap. Mixture-of-experts (MoE) routers produce a per-token distribution over experts. Prior work studies router uncertainty to improve MoE fine-tuning [15], and, in the sources we read, does not use it as a signal for whether an answer is correct or should be escalated.

Our contribution. We test the hypothesis that expert-routing statistics (normalized router entropy, its change across layers, and expert-selection instability) predict small-tier correctness. The small tier would be a small open MoE model (candidates such as OLMoE [16], to be confirmed in a feasibility spike for MLX support and router access). The signal is available at prefill and at decode time at near-zero extra cost, which makes the skip rule and early abort of C1 cheaper to apply. It is compared on AUROC and calibration error against log-prob, margin, self-consistency, and a learned hidden-state probe (our fallback if no suitable MoE model runs locally). A negative result is a reportable outcome.

3.3 C3: Calibrated routing with guarantees

Established. Conformal Cascade [9] defers when a conformal prediction set is not a singleton, with a coverage guarantee (shown on multiple-choice benchmarks; open-ended generation left for future work). RouteNLP [10] gives a marginal conformal guarantee on a violation probability. RACER [11] gives distribution-free risk control for set-valued routing. Conformal risk control [12] is the general tool.

Gap. We have not found a guarantee that covers the full *tax-aware* policy (skip, early abort and gate together), and generation tasks (math word problems, code with unit tests) are less covered than multiple-choice ones.

Our contribution. Let $A(\lambda)$ be the event that the small answer is accepted under threshold λ after any skip or abort rule, and define the loss $\ell(\lambda) = \mathbf{1}[A(\lambda) \wedge S \text{ wrong} \wedge L \text{ right}]$, which is non-increasing in λ and zero when the small answer is not accepted. Since the error of the cascade minus the error of always-large is at most ℓ, choosing $\hat{\lambda}$ by conformal risk control on a calibration split gives $\mathbb{E}[\ell(\hat{\lambda})] \leq \alpha$, i.e. an expected quality loss relative to always-large of at most α. The guarantee is marginal (not per prompt), assumes exchangeability, and weakens under distribution shift; we test coverage empirically with repeated random splits and under a deliberate shift (calibrating on one dataset and testing on another).

3.4 C4: Per-step routing in agentic workflows

Established. Budget-aware per-step routing for multi-step agents, trained with boundary-guided reinforcement learning, reduces dollar cost on agent benchmarks [13]; its cost is token dollars and router overhead is excluded from the reported metrics.

Gap. In an agent loop the context grows each step, so moving a step from the local tier to the hosted tier re-sends the whole history (the hosted model must prefill it again), and each tier switch adds latency. We have not found this switching cost modeled or measured on a local-plus-hosted pool.

Our contribution. We extend the C1 model to a trajectory: the cost of escalating at step t includes the wasted small-model step and a prefill term that grows with context length, and a policy chooses the tier at each step. We evaluate on a lightweight multi-step tool-use benchmark that fits a laptop (candidates: function-calling data such as xLAM, or a small multi-step subset of an agent benchmark; feasibility is checked in a spike). We compare per-step policies (always-small, always-large, per-step threshold from C3, a learned step router in the style of [13]) on task success, dollars, seconds and joules.

3.5 Priority and fallback

C1 and C3 are the core, because they build directly on the measurement pipeline. C2 and C4 carry more risk. If no MoE model runs locally, C2 falls back to a hidden-state probe; if no suitable agent benchmark runs on the laptop, C4 is restricted to multi-step chains assembled from the same datasets. Whether C2–C4 beat the baselines is open; null results will be reported.

Contribution	Closest prior work	Open gap addressed
C1 Escalation cost	[1, 7, 10]	Latency, tail, energy on local+hosted pool; tax-aware policies
C2 Confidence signal	[8, 19, 5, 15]	MoE router statistics as escalation signal
C3 Guarantees	[9, 10, 11, 12]	Guarantee for skip/abort policies; generation tasks
C4 Agentic per-step	[13, 14]	Switching/re-prefill cost across steps; measured latency and energy

4 Architecture

Client layer (dashboard, offline harness) → FastAPI router service (ingestion and cache, feature extractor, policy engine, confidence-signal module, conformal calibrator, cost and latency accountant, and a step controller for agent loops) → tiers (small and medium MLX-LM 4-bit local, large hosted API) → trace store (SQLite/Parquet). Evaluation is trace-driven: outputs, signals, log-probs, timings and energy are cached per prompt (or per step) and tier, policies are replayed offline, and a live subset validates replay fidelity. Caching every tier's output for every prompt is also what makes the counterfactual losses in C3 computable.

5 Two Paths

Path A (local, medium tier) Small and medium local tiers; routing between them; baselines; proof of concept; signal study (C2) on local models; started in CMPE 295A.

Path B (large tier) Adds the hosted large tier and real cost accounting; the guarantee experiments (C3), tax-aware policies and agentic extension (C4); completed in CMPE 295B.

Both paths share the same code, trace format and harness.

6 Features

- Pluggable policies and baselines (always-S/M/L, random, feature router, cascades, cascade routing, oracle, P3, P4, per-step policies).
- Confidence signals: mean log-prob, minimum token probability, margin, self-consistency, self-verification, MoE router statistics, hidden-state probe.
- Conformal calibrator that outputs a threshold and its stated guarantee for a chosen α.
- Per-request and per-step accounting of cost, TTFT/TPOT/end-to-end time and energy, split into useful and wasted work.
- Trace store, replay harness, statistics (bootstrap intervals, paired tests), frontier plots, and a web dashboard.

7 Requirements

- Hardware: Apple M4 MacBook (variant and RAM recorded in setup). Hosted large-tier API with a small spending cap.
- Software: MLX-LM, llama.cpp/Ollama (fallback), FastAPI, SQLite, Parquet, optional Redis, a sandbox for HumanEval.
- Models: Qwen2.5 small ≈1.5B, medium 7B (4-bit), large 72B hosted [20]; a small open MoE model for C2; Llama 3.x fallback.
- Data: ARC-Easy, MMLU, GSM8K, HumanEval, about 1,400–1,500 stratified prompts split 40/20/40 into router-training, calibration and test sets; a multi-step tool-use set for C4.

8 Evaluation Plan

E1 per-tier characterization; E2 signal quality (AUROC, calibration error) for all signals including C2; E3 tax measurement; E4 policy frontiers; E5 sensitivity to ρ; E6 tail latency; E7 replay-versus-live fidelity; E8 guarantee coverage (violation rate against α over repeated splits and under dataset shift); E9 agentic per-step policies (task success, cost, time, energy, switching overhead). Quality must be within ϵ (2–5 points) of always-large. Dollars, joules and seconds are reported separately. Threats: thermal throttling, benchmark contamination, API variance, replay fidelity, small calibration sets. No results exist yet.

9 Timeline (two semesters)

Weeks	Milestone
295A 1–3	Abstract, formation presentation, topic and advisor sign-off
295A 3–9	Literature and baselines; feasibility spikes (log-probs, hosted tier, energy, external router code, MoE model access, agent benchmark); architecture and design
295A 9–15	Functional specifications; Path A proof of concept; first signal study (C2); report chapters 1–2
295B 1–7	Path A complete; trace collection; hosted tier integration; tax measurement (C1)
295B 5–10	Tax-aware policies P3/P4; conformal calibration and coverage experiments (C3)
295B 8–12	Agentic per-step extension (C4); full evaluation E1–E9
295B 11–14	Testing, deployment, performance analysis; report; advisor reviews
295B 14–15	Presentation, poster, Project Exposition, final report

10 Deliverables

Router service; trace dataset; evaluation harness; tax-aware policies; confidence-signal library including the MoE-router signal; conformal calibrator; agentic step router; dashboard; analysis report; paper draft; open-source release.

11 Risks

- Log-probs or MoE router outputs unavailable in MLX-LM: feasibility spikes; fall back to llama.cpp/Ollama, or to a hidden-state probe for C2.
- Hosted API price or availability changes: cache all large outputs; two providers.
- Thermal throttling and API variance distort latency: repetition, logging, cooldown.
- Overlap with fast-moving related work (conformal routing, agentic routing): re-check the literature monthly and narrow the claims if a gap closes.

- Scope: four contributions in two semesters is ambitious; C2 and C4 are the first to reduce to a negative or partial result.
- The tax proves small or routers equal cascades: report it as a finding.

12 Expected Outcomes

A measured account of when the cost of escalation makes cascading worse than routing in cost, latency and tail latency on commodity hardware; an evaluation of whether MoE router statistics are a useful escalation signal; thresholds with a stated, empirically tested quality guarantee for tax-aware policies; a first measurement of tier-switching cost in agent loops; and an open, reproducible system.

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